System Description of ShanHe-Huawei_OceanStor_A600_1AMDClient-1A600

1. Basic Information

1.1 System name of the submission

ShanHe-Huawei_OceanStor_A600_1AMDClient-1A600

1.2 Submitter name

JNIST

1.3 Submission date

July-10-2025

1.4 Version of the benchmark

MLPerf Storage V2.0

1.5 Solution type of the submission

Proprietary - OceanStorOS

1.6 Submission division (OPEN or CLOSED)

CLOSED

1.7 Power Requirements

System component	Power supply unit	Nameplate rated power	Design Power
A600 Appliance 1	Power supply 1	3000W	3000W
	Power supply 2	3000W	
Totals		6000W	3000W

1.8 System Topology

The test environment consists of one host and one Huawei OceanStor A600, both connected to an ethernet RoCE switch.



1.9 Rack Units Requirements

The Storage system consumes 2U space.

2. Hardware

2.1 Host

The host node is a commercial server with 2 AMD EPYC 9654 96-Core Processors. The Client communicate with the storage through RoCE. The hardware details are as follows:

Component	Description	Count
CPU	AMD EPYC 9654 96-Core	2
Memory	DDR5 4800MHz 64GB	24
Network Card	Mellanox MT28908 ConnectX-6	4

2.2 Storage

The Huawei OceanStor A600 is a new generation of high-performance AI storage specifically designed for AI, offering ultimate performance to accelerate large model training and inference.
Based on a new control-data separation architecture, Huawei OceanStor A600 high-performance AI storage supports flexible expansion with up to 32 controllers, matching multi-modal large models with trillions to quadrillions of parameters. It smoothly evolves to meet the demands of AI-native vector, tensor, and KV Cache data paradigms. It can be equipped with a high-performance vector retrieval engine to accelerate large-capacity vector knowledge base retrieval, reduce inference "hallucinations," and provide multi-level KV Cache capabilities to reduce inference latency and improve the cost-effectiveness of inference. This storage system meets the full-process data processing needs of AI training and inference.



The hardware details of one appliance are as follows:

Component	Description	Count
CPU	HiSilicon Kunpeng-920	4
Memory	DDR5 32GB	32
Network Card	2*100G RoCE	12
Drive	NVMe SSD 7.68TB	36

2.3 Switch

Huawei CloudEngine 9860 switch is used. The host and storage are connected to it.

3 Software

3.1 Host

The configuration of the host software is shown as follows:

Software	Description
OS	openEuler release 22.03 (LTS-SP4)
Kernel	5.10.0-216.0.0.115.oe2203sp4.x86_64
Mellanox OFED version	MLNX_OFED_LINUX-5.8-1.1.2.1

3.2 Storage

The software deployed on the storage is OceanStor A600 V700R001C10.